

Feature Review

Beyond punishment: psychological foundations of restorative interventions

Katherine McAuliffe ^{1,*}, Julia Marshall^{1,2}, and Abby McLaughlin¹

Work on the psychology of justice has largely focused on punishment. However, punishment is not our only strategy for dealing with conflict. Rather, emerging work suggests that people often respond to transgressions by compensating victims, involving third-party mediators, and engaging in forgiveness. These responses are linked in that they are involved in more restorative than retributive justice practices: they center victims as well as (or instead of) perpetrators and can help repair fractured relationships. Work with non-human animals echoes these findings: reconciliation and intervention by third parties play a key role in conflict management across taxa. In this review, we focus on these restorative interventions, with the aim of painting a more complete picture of the psychology of justice.

Coping with conflict

Transgressions crop up across all realms of social life, from intimate interpersonal relationships – a close friend makes a joke at your expense at a party – to large, impersonal institutions – a failing company defrauds naïve investors. Transgressions like these represent an ever-present threat to flourishing social relationships, which are crucial to our success as individuals and as societies. To cope with this threat, humans and non-human animals employ a range of response strategies in the face of conflict. Strategies for responding to transgressions form the foundation of systems of justice.

One type of justice-related response that has received a great deal of attention across disciplines is **punishment** (see [Glossary](#)). Punishment involves the imposition of a cost on a transgressor by an agent, who often incurs a cost to punish [1,2]. While a plurality of motives likely support punitive responses, including deterrence and **retribution** [3], communication [4], and competition [5], these motives are aligned in that, generally speaking, they are directed toward perpetrators of wrongdoing rather than victims.

Our perception of punishment as a – if not the – main justice-related response is connected to the fact that, in many empirical studies, the only way to respond to a transgression is to punish, and to the fact that, in Western societies, punishment is the most common response to crime [6]. This keen interest in punishment has deep roots in studies of human behavior and society (e.g., [7]). However, recent years have seen a rapid expansion in research on the kinds of justice-related responses people pursue both under natural conditions and when they are given more options in experimental paradigms. This expansion has focused on non-punitive responses that have connections to **restorative justice**. Unlike punishment, restorative interventions clearly center victims in the justice process. Here, we begin by discussing limitations of the focus on punishment before moving on to review three kinds of restorative interventions that have received recent empirical attention from psychologists, economists, and anthropologists: (i) **victim compensation**, (ii) third-party **mediation**, and (iii) **forgiveness** by victims. We then (iv) contextualize these findings in comparative work on animals' responses to transgression, which show that behaviors relating to **reconciliation** from both victims and third parties are common and taxonomically widespread. Together, this work on restorative interventions complements existing

Highlights

Strategies for coping with conflict are important for restoring and sustaining valuable social relationships.

Punishment has garnered much attention as an intervention against transgressions, particularly in terms of its role in promoting cooperation and as a pillar of Western justice systems.

Yet, a surge of recent work has found that non-punitive interventions, such as victim compensation, third-party mediation, and forgiveness are also important strategies for responding to transgressions.

Work with adults, children, and across diverse societies highlights that these restorative responses – which center victims in the justice process – are early-emerging and privileged across many cultures.

Comparative work provides additional context, showing that reconciliation and intervention from third parties is taxonomically widespread and central to conflict management in animal societies.

¹Department of Psychology and Neuroscience, Boston College, Chestnut Hill, MA, 02467, USA

²Department of Cognitive and Psychological Sciences, Brown University, Providence, RI, 02912, USA

*Correspondence: mcaulikg@bc.edu (K. McAuliffe).



work on punishment, helping provide a more complete picture of the constellation of justice-related responses that help support complex social interactions.

Punishment is important, but it is not everything

One reason why punishment has received so much attention is because of its promise in helping solve the problem of cooperation. Large-scale cooperation among non-relatives is hard to explain and punishment offers a viable solution: both theoretical and empirical work show that cooperation can be sustained if non-cooperative agents are punished (e.g., [8]). Note, however, that punishment is by no means the only mechanism purported to sustain cooperation at the proximate or ultimate levels (e.g., [9,10]), nor is it clear whether punishment is particularly pronounced in the context of cooperation relative to other contexts. Moreover, although punishment can sustain cooperation within groups, this does not necessarily mean that punishment is beneficial to people at the individual level (e.g., [11]) nor that it is always beneficial to groups [12].

The empirical focus on punishment in the context of cooperation has shaped how punishment is often studied. Specifically, because the goal is to explain how large-scale cooperation among strangers is maintained, paradigms often involve one-shot, anonymous contexts and measure the extent to which people invest in punishment of non-cooperators. This approach to measuring punishment has been hugely generative. We know that punishment happens both in second-party contexts [13], in which the victim punishes the perpetrator of a transgression, and in third-party contexts [14], in which an uninvolved bystander punishes a perpetrator. We also know that, consistent with theoretical predictions [8], punishment can promote cooperation in incentivized contexts (e.g., [15,16]). Punishment is observed, to varying degrees, across diverse human societies under both experimental [17,18] and natural conditions [19–23]. Both second- and **third-party punishment** emerge early in childhood (for reviews see [24,25]). Finally, comparative work suggests that punishment helps regulate non-human animal cooperation in some contexts (for reviews see [2,26]).

A byproduct of the ways in which punishment is often studied – namely, with punishment operationalized as direct fines in anonymous contexts – is a skewed perception of both the form and frequency of punishment under natural conditions [27]. When researchers have explored punishment in daily life, they have found that direct punishment does happen, but often takes the form of verbal chastisement (e.g., scolding someone for littering; [28–30]), presumably because direct fines would be difficult, if not impossible, to instantiate under ‘daily life’ conditions. Additionally, it is often unclear who is in a legitimate position to levy fines against transgressors. Work capitalizing on non-experimental methods, such as experience sampling [31], has shown that indirect forms of punishment, like gossip, are more common than direct confrontation [32]. Moreover, punishment is influenced by relationship quality, which means that a focus on anonymous interactions constrains our inferences. For example, people report desiring punishment more when they are dealing with an abstract other than with their partner or spouse [33], and third parties are more likely to punish if they value victims’ welfare [34]. Of particular relevance to the current review: rates of punishment reported in experimental work are likely inflated because the alternative to punishment, in many experimental paradigms, is simply to do nothing [27]. Thus, any desire to intervene against transgressions is necessarily channeled into punishment.

What, then, would people choose if a broader range of options were on the table? One non-punitive response is simply to abandon the transgressor, cutting off the relationship entirely (**partner rejection**; Box 1). However, in other situations, responses that restore justice to victims and make space for future positive social interactions are preferred. The following sections focus on these more restorative interventions. Before moving to these sections, however, we wish to make two disclaimers. First, the majority of studies reviewed in the following sections focus on

Glossary

Arbitration: involvement of a third-party in conflict resolution whereby the arbiter delivers a binding decision that the disputants must follow.

Forgiveness: an internal shift in which the victim releases negative feelings and moves toward positive feelings following a transgression.

Mediation: involvement of a third-party in a conflict with the goal of reaching a (usually) non-binding agreement.

Partner rejection: decision to cease interacting with a social partner following a transgression.

Policing: a behavior related to punishment in which an agent inflicts a cost on a transgressor and, in doing so, generates public goods benefits.

Punishment: the imposition of a cost on a transgressor by an agent.

Reconciliation: repairing a relationship following a transgression.

Restorative justice: an approach to justice that de-emphasizes punishment of transgressors and emphasizes interventions that promote positive outcomes for victims.

Retributive justice: an approach to justice that emphasizes the deserved suffering of transgressors as a means to reciprocate the harm transgressors have caused.

Sanction: a behavior related to punishment in which an agent inflicts a cost on a transgressor and immediately benefits.

Second-party punishment: punishment by the victim of a transgression.

Third-party punishment: punishment by a bystander who is not, themselves, involved directly in a transgression.

Victim compensation: help directed toward the victim of a transgression.

Box 1. Partner rejection

Instead of deciding to punish or restore relationships, people can choose to reject partners – that is, cease interactions with them, which is considered a form of partner choice [171]. Empirical evidence from adults, children, and non-human animals suggests that partner rejection is not an extreme form of punishment but, rather, is a distinct response (see [172] for a review). For instance, partner rejection shows enhanced sensitivity to intent relative to punishment decisions, which are particularly sensitive to outcomes [173]. This difference is echoed in work exploring the developmental emergence of sensitivity to intent in children's punishment versus partner rejection decisions [174]. Specifically, younger children tend to weigh outcomes over intent in their punishment decisions, recommending more punishment for accidental (i.e., unintentional) harm than attempted (i.e., intentional but unsuccessful) harm, whereas older children show the reverse pattern in their punishment decisions [124,174]. Underscoring partner rejection's sensitivity to intent: children show intent-based partner rejection decisions before they show intent-based punishment decisions (Figure I), suggesting that intent information is privileged for partner rejection in development. Other new work has explored punishment versus partner rejection in an intergroup context, finding that second-party punishment depends on the group membership of perpetrators (more punitive of out-group relative to in-group transgressors), whereas partner rejection does not [175]. Again, this finding underscores partner rejection as a distinct response to transgression – one that is dissociable from punishment. Finally, comparative work suggests that partner choice is relatively widespread taxonomically, at least in the primate lineage, whereas punishment is relatively taxonomically constrained, perhaps due to the cognitive demands of punishment relative to partner choice [176].

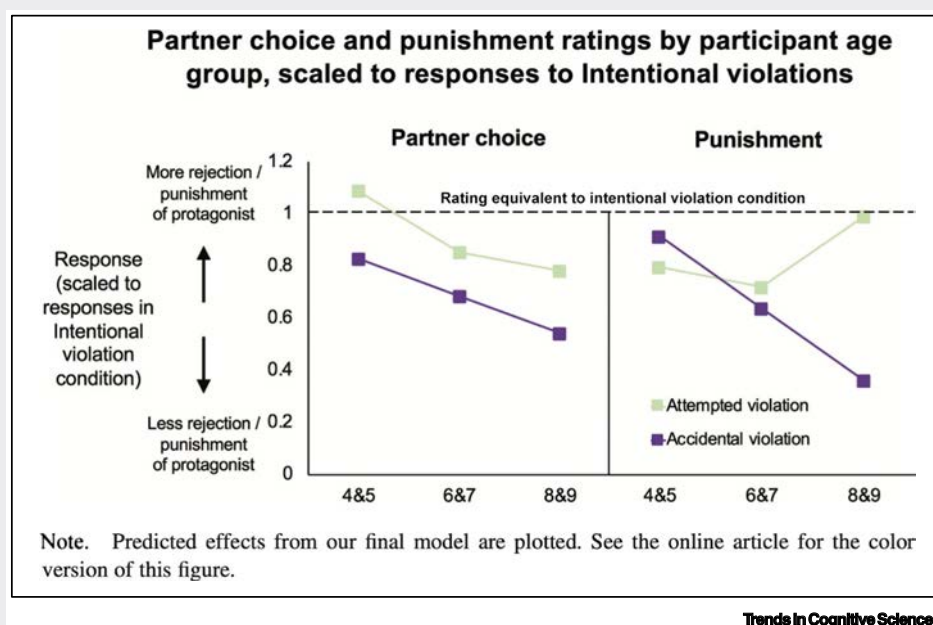


Figure I. Partner rejection decisions are sensitive to intent earlier in development than are punishment decisions. From Martin *et al.* (2022) [174].

Western educated industrialized rich and democratic (WEIRD) populations, reflecting the well-established WEIRD-bias in psychological studies [35]. We have noted cases in which a study focused on a non-WEIRD population. Second, while we are focusing on justice in terms of responses to transgressions, we do not mean to imply that this represents a holistic view of justice. A large literature has explored theories of justice as fairness – particularly in terms of distributive fairness – in both humans and non-human animals. Here, however, our focus is specifically on theories of justice as they related to responses to transgressions and relationship repair.

Victim compensation

Economic tasks that have provided participants with the option to restore justice to victims through compensation and/or to punish perpetrators find that both second and third parties

often opt for compensation [36] (see [37] for a review). Compensation in these cases typically involves directing resources to the victim of transgressions, but in the real-world can also involve providing aid or comfort to victims [38] or acknowledgments of wrongdoing [39].

Victim compensation in adults

Experimental work with adults focusing on fairness norm violations found that third-party observers would punish unfair perpetrators when punishment was the only option [40]. However, when both punishment and victim compensation were options, they were more likely to compensate victims. Related work using a task in which participants could invest continuous amounts of money in compensation and punishment found that participants invested more in compensation [41]. Similarly, when punishment and compensation were pitted directly against one another in a second-party fairness task, people overwhelmingly preferred compensation to punishment [42].

While a robust literature suggests that people generally prefer compensation to punishment [36,37], other work suggests that punishment may be preferred under some conditions and compensation under others. For instance, a recent study explored participants' investment in only punishment versus only compensation (versus both) in a task that assessed responses to unfair resource allocations from both second- and third-party perspectives [42]. They found that victims of unfairness (i.e., second parties) often preferred compensation, whereas third-party observers of unfairness preferred an option that resulted in both compensation of the victim and punishment of the perpetrator (in the case of extremely unfair offers). In contrast, other work using a slightly different paradigm found that participants were more likely to punish than to compensate, and invested more in punishment as second parties than as third parties [43], suggesting that these preferences may be situationally-dependent.

Research exploring the proximate mechanisms supporting decisions to punish versus compensate has pointed to roles of intent, empathy, and moral outrage: (i) intent appears to influence third-party punishment but not compensation [44], (ii) people who experience more empathy are more likely to engage in compensation [45,46], (iii) compensation is predicted by moral outrage at both the trait and state level [47].

An important area for future work is developing a better understanding of the complex relationship between punishment and compensation, as they likely both feed into the psychology of justice in different, yet related, ways. Some argue that punishment satisfies people's desire for justice, making them less concerned with victim compensation [48]. However, compensation may decrease people's desire to seek punishment: when participants were presented with scenarios in which victims expressed satisfaction with a restorative outcome, they endorsed the pursuit of milder punishment [49]. Moreover, compensation decisions, in terms of the amount conferred to victims, have been shown to be tethered to the degree to which perpetrators are punished [50]. Finally, although there exists some work tackling the reputational benefits associated with compensation (Box 2), more work is needed to understand whether reputational concerns are a major driver of decisions to compensate victims. Nevertheless, the findings that exist to date speak to a complex interplay between punishment and compensation decisions in adults, which future work should seek to better understand.

Victim compensation in children

From early in development, children show intuitions about when punishment should be meted out in response to transgressions and engage in punishment and related interventions themselves, (e.g., tattling, protesting; see [25] for a review). In general, work on punishment in children has explored whether children punish when the alternative is to do nothing (e.g., [51]). Given this,

Box 2. Reputation of helpers and forgivers

People can gain reputational benefits by punishing transgressors, particularly as uninvolved third parties. Indeed, reputational benefits provide an attractive explanation for why costly third-party punishment is worth the investment [177]. In line with the reputational account, people who engage in punishment as third parties are judged as more trustworthy than those who do not punish [178]. Moreover, these judgments reflect reality: those who are willing to punish inequality behave in more trustworthy ways in economic games [178]. However, when third parties are given the opportunity to help victims in addition to punishing perpetrators, the reputational benefits of punishment are attenuated [178], suggesting that helping may be a more powerful signal than punishment.

Other work shows that third-party punishers are rewarded for intervening (more so than those who do nothing), but third-party helpers are rewarded even more than punishers [177]. In this same vein, recent work found that, across many different paradigms, individuals who responded to injustice via compensation rather than punishment accrued greater reputational benefits [179]. Similar effects have been found when measuring people's willingness to vote for different candidates: candidates who compensate victims are preferred over those who punish perpetrators [180]. Interestingly, preferences for those who punish versus those who compensate may depend on whether the agent responding to a transgression is a victim themselves (i.e., a second party) or a third-party observer. Work has also shown that people prefer victims who respond to transgressions prosocially (e.g., through compensation) to those who punish [181]. Similarly, people prefer third parties who respond prosocially to those who punish, yet punishing third parties are preferred over punishing victims.

Beyond compensation, there is some work finding that children and adults evaluate forgiveness positively. For example, adults negatively evaluate victims who refuse to forgive in cases in which a perpetrator has attempted to reconcile, implying that they hold an expectation that victims will engage in forgiveness [115], and other work has found that victims who forgive mild offenses are viewed as warmer than those who withhold forgiveness [182]. Similarly, work has shown that children, like adults, show a preference for an agent who chooses to forgive a transgression over an agent who chooses not to forgive [127]. Although these studies did not tackle reputation *per se*, they suggest that the groundwork may exist for forgiveness to carry reputational benefits in the same way that helping victims of a transgression does.

inferences from developmental work on punishment have principally centered on the contexts in which children are willing to punish perpetrators and the motivations supporting these decisions [52–54]. While these findings suggest that children have an appetite for punishing perpetrators, they do not rule out the possibility that children are also motivated to intervene in ways that benefit victims. For instance, recent work suggests that children in the USA and China punish as third parties in a way that helps achieve equality between a selfish perpetrator and their victim [54,55]. Additionally, third-party punishment in children tested in China has been shown to be sensitive to victim distress [56]. These data help set the stage for work that has directly explored victim compensation more directly in children

How do children weigh concerns for punishing perpetrators versus compensating victims? In a recent behavioral study [57], 6- to 9-year-olds were third-party decision-makers in a task in which they could intervene against unfair resource allocations to either punish selfish perpetrators (remove all their resources), compensate victims of selfishness (ensure that perpetrator and victim had the same number of resources), or do nothing. While children engaged in both punishment and compensation, only punishment was selectively directed toward unfairness, whereas compensation happened in response to both unfair and fair allocations. These findings suggest that children preferentially intervene against unfairness with punishment and not victim compensation. Broadly in line with these results, developmental work shows that children endorse punishment of transgressors, even though they find compensating victims to be more rewarding [58].

In tension with the finding that children prefer punishment to compensation, however, is work examining children's evaluations of punishment versus compensation by third parties. In a study that presented 5- to 9-year-old children with scenarios in which third parties either punished unfair agents or compensated victims of unfairness [59], children showed a preference for acts of compensation relative to acts of punishment, though they evaluated both kinds of intervention positively. This preference translated to a preference for the interveners themselves: compensators were preferred over punishers.

Of course, in the real world, people do not typically have to choose to either punish or compensate. And, indeed, sometimes restoring justice to victims can also involve inflicting costs on perpetrators. In a study with 3- to 5-year-olds, participants could respond to a thief who stole an object from a victim either by punishing the thief (removing the stolen item) or restoring justice to the victim (removing the stolen item and giving it back to the victim) [60]. Perhaps not surprisingly, children tended to prefer the restorative option (consisting of both compensation and resource removal) rather than the option that resulted in punishment alone. This finding was recently replicated in a sample of 3- and 5-year-olds in China [61]. Indeed, by the age of 5 years, Chinese children appear to opt for restorative justice, consisting of both punishment and restoration, both when they are the victims of transgressions (i.e., in a second-party context) as well as when they are uninvolved bystanders (i.e., as third parties) [62]. Moreover, Chinese children assign more positive evaluations to victims who take restorative rather than strictly punitive measures in response to a transgression [63].

Although work that assesses children's preferences for one intervention over the other can be helpful in revealing which motives are privileged (e.g., are children principally concerned with harming perpetrators or helping victims?), in actuality children are likely adjudicating between response strategies in a more continuous, rather than dichotomous, manner. Consequently, a productive line of inquiry is trying to understand how these different response strategies are calibrated across different social contexts, transgression types, and ages. For instance, some work suggests that older children in China are more likely to endorse restorative measures over punishment relative to their younger counterparts [64] and that information about intent [65] and contextual cues about perpetrators and victims (e.g., status) inform the decisions of children in Canada and the USA about whether and how much to punish versus compensate [66].

Taken together, the studies reviewed earlier suggest that adults and children often opt for victim compensation either instead of or in addition to punitive responses. However, when both punishment and compensation are on the table, punishment alone is rarely chosen. These findings suggest that people take victim needs into account following transgressions and use compensation as one means of restoring justice.

Third-party intervention, arbitration, and mediation

When conflict arises in daily life, intervention from third parties can play an important role in interrupting it and guiding disputants toward a resolution. A study of video footage from public spaces in Amsterdam revealed that third parties regularly involved themselves in others' conflicts [67]. Moreover, when third parties intervened with physical force, their intervention often disrupted conflicts, pointing to the immediate, proximate-level benefits of third-party intervention. In the Peacemaker Game, a game in which third parties can intervene to alter payoffs from competitive to cooperative, even the possibility of intervention by third parties is sufficient to promote cooperation [68], pointing to the adaptive value of third-party intervention – at an ultimate level – as a mechanism for supporting cooperative relationships.

Third-party mediation occurs when a third party is involved in helping resolve a conflict between adversaries [69]. In the legal literature, a distinction is drawn between third-party mediation and third-party **arbitration** [70]: in the case of mediation, the third party can suggest a solution, but the disputing parties are not required to adopt it, while in the case of arbitration, the third party can enforce the solution.

Third-party arbitration can be influential in cases in which parties are uncertain about (or have different views on) whether a transgression has occurred. For instance, in a simulated cooperative

dilemma in which information asymmetries exist between players (i.e., about whether players have cooperated or defected), third-party arbitration – in which a third party can correct misunderstandings about past behavior – helps sustain cooperation [71].

For our purposes, we subsume third-party arbitration under the umbrella term ‘third-party mediation’ to refer to cases in which a third party is invited by disputants into the justice process. Mediation is effective across an impressive range of contexts: between peers [72], within families [73,74], between hospitals and their patients [75], in the context of environmental disputes [76], and even in the context of international conflicts [77,78].

Third-party mediation in adults

Recent work in anthropology underscores the role that third-party mediation plays in coping with conflict. A study exploring justice responses in the Enga of Papua New Guinea found that third parties were often involved in mediating conflicts. Moreover, justice responses resulting from this mediation process, across a range of transgressions, were principally oriented toward the restoration of damaged relationships. Mediation could, for example, result in the transgressor agreeing to give resources to the victim (i.e., both punitive and compensatory outcomes) or instructions to the transgressor to change their behavior. While punishment from the third parties themselves was rare, victim compensation was common. Thus, justice among the Enga appears to be principally concerned with victim rights as opposed to perpetrator behavioral reform or retribution [79]. In a similar vein, a recent study examined transgressions among the Mentawai of Indonesia [80]. There was no evidence that third parties punished transgressors directly. Rather, third parties, who were typically either government officials or high-status community members, mediated conflicts and helped facilitate reconciliation. Oftentimes, reconciliation was achieved through the payment of Tulou, a form of compensation from the perpetrator and/or the perpetrator’s family to the victim and/or the victim’s family. These findings suggest that justice-related interventions principally help restore relationships as opposed to enforcing norms. Ethnographic records from Kiowa equestrian foragers from North America and Nuer pastoralists from South Sudan generate a similar picture: third parties intervene to resolve conflicts, foster forgiveness, and restore relationships [81]. In general, then, available evidence from anthropology suggests that justice in these contexts involves a balance of victim needs and perpetrator punishment rather than being specifically focused on promoting cooperation through targeting perpetrators.

Unlike experimental work, which typically involves bystanders, perpetrators, and victims with no pre-existing relationship (e.g., [14]), third-party mediation in human societies typically involves parties with established roles or relationships. Who, then, mediates conflict? Mediators are often people in leadership roles, which presumably lends legitimacy to their involvement in others’ conflicts [82]. Indeed, work drawing on ethnographic records suggests that a principle function of leaders in human societies is to help manage conflicts [83]. In other cases, they are appointed magistrates [79], government officials [80], or respected community members [80]. A recent paper exploring third-party mediation based on self or peer reports from pastoralists in Ethiopia highlights the important role of third-party mediators in regulating social relationships [84]. They find that mediation is more common in more severe conflicts and that gender dynamics influence the kinds of mediation that are applied to different conflicts.

There exists relatively little recent experimental work on the effects of mediation on the outcomes of conflict, yet older work suggests that mediation successfully resolves disputes [85]. More recent experimental work suggests that socio-cultural forces may shape people’s inclination to call on mediators: in an experimental set-up that created a conflict between a participant and a confederate partner, participants in Turkey were more likely to call on third parties to help resolve

a dispute than were participants in the USA. Additionally, and in line with research on victim compensation, empathy may play a proximate role in guiding third-party mediation of conflicts toward resolution [86]. Perspective-taking by disputants may also facilitate mediation efforts [87,88].

Third-party mediation in children

From a developmental perspective, we know that children often call on third parties to help manage conflict [25,89], yet we know relatively little about how mediation skills develop in childhood. A recent study in Colombia suggests that spontaneous third-party mediation seldom occurs in children and early adolescents, but it can be trained [90]. Work with Japanese preschoolers suggest that adults help scaffold the development of mediation behavior through modeling language that serves in mediating conflict [91]. Finally, a large body of work points to the promise of peer mediation programs in schools as a fruitful area for mitigating peer-to-peer conflict (e.g., [72,92]). Work in this area is also exploring the role of robots in serving as third-party mediators for both children and adults alike [93,94].

Together, this work suggests that people often rely on third parties in justice proceedings. Third-party mediators can help restore justice through both attending to victim needs following a transgression as well as by making sure that perpetrators are held accountable for their misdeeds. Third-party mediation, perhaps because it gives agency to all parties in the justice proceeding, can be a successful means of repairing relationships damaged during conflict.

Forgiveness

Although there are many definitions of forgiveness, one that has been influential in shaping research on forgiveness as a response to transgressions is its conceptualization as ‘a suite of prosocial motivational changes that occurs after a person has incurred a transgression’ [95]. With this operationalization in mind, there are two key points to make. First, while forgiveness can be an important step on the path to reconciliation, it is not the same as reconciliation. Second, it is clear that while other parties can play a role in processes that promote forgiveness and reconciliation (e.g., through third-party mediation), forgiveness itself must be done by the victim.

One notable feature of the study of forgiveness is that it spans an especially wide range of disciplines and methodological approaches (Box 3). A set of recent reviews from across different fields helps showcase this interdisciplinary breadth: there have been advances in work on forgiveness across neuroscientific [96], evolutionary [97], organizational [98], developmental [99–101], philosophical [102,103], theological [104], and behavioral medical [105] approaches. These reviews highlight the recent progress that has been made in developing a fuller understanding of forgiveness and point to its potentially important role in regulating social relationships. Work at the ultimate level of analysis suggests that forgiveness can help stabilize cooperation and maintain valuable relationships (e.g., [190]). For example, forgiving strategies in Prisoner’s Dilemma games can promote cooperation [106]. Forgiveness may also confer reputational benefits (Box 2), suggesting that there may be multiple selective pressures acting on its evolution. Here, we are specifically interested in the psychology of forgiveness by victims and how it is instantiated in response to transgressions.

Forgiveness in adults

Firstly, work with adults shows that forgiveness is sensitive to proximate-level inputs. For instance, forgiveness is influenced by information about intent: people are more likely to forgive unintentional acts than intentional acts [107]. Forgiveness is also influenced by empathy [95,108], and perspective-taking [109]. A second advance is work showing that forgiveness is, to some extent, dependent on expressions of remorse, such as apologies [110]. Apologies communicate to

Box 3. Methods for studying forgiveness

Unlike victim compensation and third-party mediation, forgiveness is an internal (rather than external) process. Because forgiveness occurs internally, there are unique methodological challenges in firmly establishing when, where, and why it is happening (Table I). As such, empirical research on forgiveness has relied principally on self-report measures and recall studies (e.g., [183–187]), both of which are language-dependent. This means that these methods are useful for studying forgiveness in adults, but less so in young children due to limitations in verbal abilities.

Other work has aimed to measure forgiveness at the behavioral level; for instance, by exploring when people are willing to resume cooperation with those who have previously defected in a cooperative game [106,183], or, in developmental work, conferring a benefit (giving cloth flowers or craft supplies) to a previous transgressor [122,126]. Behavioral measures are useful in that they can potentially track the consequences of forgiveness on future interactions between a transgressor and victim. However, they are limited in so far as they do not necessarily measure forgiveness. For instance, strategic motives may lead to cooperation with a previous transgressor regardless of whether the victim has truly forgiven the transgressor, and victims may allocate resources to apologetic transgressors because they perceive them as nice, not because they have forgiven them. Finally, forgiveness is sometimes measured as the absence of punishment (e.g., [188,189]). Like the behavioral markers described earlier, the lack of punishment could stem from forgiveness (and, indeed, some scholars operationalize forgiveness as the opposite of revenge [190]). However, as with behavioral measures, the lack of punishment does not necessarily indicate that forgiveness has happened (e.g., fear of retaliation may prevent a victim from punishing but would not necessitate forgiveness). Also, this operationalization does not leave room for the potentially important relationship between punishment and forgiveness; namely, that punishment can promote forgiveness [191,192].

Table I. Methods for studying forgiveness^a

Method	Example implementation	Strengths	Weaknesses
Self-report	Participants answer questions about when and why they are willing to engage in forgiveness [186]	Closely aligned with how forgiveness is operationalized as an internal process Relatively straightforward to implement and, thus, scalable Provides insight into general tendency to forgive, rather than forgiveness at one time point	Potentially vulnerable to demand effects if participants think that forgiveness is valued Heavily dependent on language and therefore limited to adults and older children Dependent on potentially varying conceptions of forgiveness across age [131] and cultural context [118,119]
Recall and diary studies	Participants recall previously forgiven and unforgiven transgressions [187]	Measure instances of forgiveness in ecologically-valid ways Allows for assessing the types of transgressions that lead to forgiveness in real life	Affected by participants' interpretation of events Susceptible to demand effects Dependent on language and therefore limited beyond adults and older children
Experience sampling	Participants asked to report conflict in relationship and then were asked questions about forgiveness at 3 subsequent timepoints [185]	Assess changes in attitudes toward transgressors in real time Measure responses to events perceived to be transgressions by participants	Can only occur within limited time range Cannot measure long-term changes in attitudes toward transgressors Dependent on language and therefore limited beyond adults and older children
Evaluative and preference measures	Comparing evaluations between transgressors (e.g., one apologized, the other did not [126]) and agents who forgave or did not [127]	Allows for direct comparisons Works with young children Assess children's ability to incorporate information about remorse, apologies, and intent	Might measure general preference – not necessarily related to forgiveness Often does not measure initial response to transgression, making it difficult to conclude that act was hurtful to victim Forced-choice questions do not allow insight into strength of attitudes and preferences
Behavioral measures	Allocating positive resources to a transgressor [122,126] or resumption of cooperation following a transgression in economic games [106].	Not dependent on language and often observable, making it suitable for young children and cross-cultural work Examines real-world consequences of forgiveness	May not be actually measuring forgiveness as other processes could drive the same behavioral outputs

^aThis table highlights the myriad ways in which forgiveness is assessed and highlights the trade-offs involved in different approaches by focusing on the strengths and weaknesses of different methods.

victims information about the importance of relationships [111], and apologies that are personally costly (e.g., canceling existing plans to make time to apologize) to transgressors may be particularly impactful in encouraging forgiveness in victims [112]. Third, and related to the role of relationship value in guiding decisions to forgive, some work suggests that apologies are particularly impactful in cases where the victim does not already have a strong relationship with the transgressor, suggesting that either a strong relationship or an apology must be in place for forgiveness to occur [113]. Fourth, a suite of recent studies has explored the consequences of forgiveness. Forgiveness can discourage future offenses in transgressors [114], and acts involving transgressors expressing remorse and victims granting forgiveness can promote reintegration into social groups [115]. However, the downstream consequences of forgiveness depend, to some extent, on how the transgressor perceives the victim's motives: when a victim is perceived as expressing forgiveness for competitive reasons, such as making the transgressor feel guilty, this may escalate the conflict, whereas when a victim is perceived as expressing forgiveness for prosocial reasons, such as desiring relationship repair, this may promote reconciliation [116]. Fifth and finally, a small but growing body of work has begun to explore how cultural ecology shapes forgiveness in adults (e.g., [117]), suggesting that conceptions of forgiveness show variation across cultural contexts [118,119]. To our knowledge, this work is being conducted principally within the field of psychology, rather than anthropology, highlighting a potentially exciting gap in our understanding of forgiveness across cultures.

Forgiveness in children

Studying the developmental roots of forgiveness is by no means a new pursuit [120]. However, there has been a recent push to better understand how forgiveness changes across development [121]. One area that has seen particular progress over the last 5 years is empirical work on forgiveness with children, much of which has employed behavioral paradigms in addition to sets of evaluation and preference questions.

Paralleling findings with adults, forgiveness in children is sensitive to transgressor intent. A recent study indexed forgiveness with a behavioral task in which forgiveness was directly pitted against punishment [122]. This study found that children were more likely to forgive than punish across conditions, a general inclination that has now been replicated across Canada, Ecuador, Peru, Uganda, and the USA [123]. This study also found that, with age, children became more likely to forgive unintentional relative to intentional transgressions. The finding that children are sensitive to intent aligns with other work showing that children are more punitive of intentional acts [124] and that 5- and 6-year-olds are partial toward an accidental versus intentional transgressor when both have displayed remorse [125].

Also in keeping with adult work, forgiveness in children is influenced by expressions of remorse. A recent study explored the conditions under which 4- and 5-year-olds would forgive a transgressor who destroyed their artwork [126]. They found that apologies were successful in promoting forgiveness (operationalized by preference in forced-choice questions and allocation of positive resources) in children as young as 4 years and that, by 5 years, children are also more likely to forgive a remorseful than unremorseful transgressor. In follow-up work, this same team found that 4- and 5-year-old children distribute more resources to forgiving victims compared with non-forgiving victims [127], suggesting that children also show a preference for forgivers.

Forgiveness in children is also sensitive to group membership. Five-year-olds are more forgiving of in-group transgressors than out-group transgressors [128,129], although remorse appears to override information about group membership. Children's sensitivity to group membership in forgiveness decisions aligns with research showing that adults are more likely to forgive those with

whom they have a close relationship [130], suggesting that forgiveness is functioning to aid in the restoration of socially valuable relationships even early in childhood.

Finally, new work explored reasoning about the consequences of forgiveness in children (aged 5 to 9 years), adolescents (aged 11 to 14 years), and adults by measuring their expectations about what positive and negative outcomes will occur after a victim forgives, punishes, or does nothing [131]. Children differentiated between the consequences of these three responses, associating forgiveness with more positive and fewer negative consequences, and associating both punishment and doing nothing with more negative and fewer positive consequences.

Taken together, insights from recent empirical work on forgiveness illustrate that adults and children readily engage in forgiveness and their propensity to do so is sensitive to a variety of relevant factors, such as intent and remorse. More broadly, work on forgiveness underscores the importance of exploring justice-related responses that can serve in repairing relationships. Forgiveness, in particular, is a response that centers the victim of transgressions as an empowered agent in the justice process. Future work in this area should continue to refine methods for capturing forgiveness at multiple levels of analysis and should explore how forgiveness relates to other justice-relevant responses.

Comparative perspectives on conflict management

As in human societies, conflict in non-human animal (hereafter animal) societies is costly. Consequently, strategies for coping with conflict confer a selective advantage. Although conflict crops up across many different social domains (e.g., foraging, mates, territory), conflict in cooperative societies has received particular attention. For instance, research has explored how free-riders – those who do not pull their weight in cooperative activities – are dealt with. Punishment originally seemed like a key candidate for answering this question [1]. Related responses include the imposition of **sanctions**, whereby the punisher immediately benefits from punishment (e.g., when a dominant individual expels a competitor from the group, thus eliminating the competitive threat) [26], and **policing**, whereby punishment generates public goods [e.g., when honeybee (*Apis mellifera*) workers kill worker-laid eggs and benefit through inclusive fitness gains] [2]. Some work suggests that the work of enforcement may be concentrated to ‘leaders’ in mammalian societies, individuals that disproportionately bear the costs of enforcing cooperation and resolving within- and between-group conflict (for a review see [132]).

Despite the promise of punishment in helping solve the puzzle of animal cooperation, empirical work suggests that punishment in animal societies is quite rare or, at least, much less common than in humans [26]. In this final section of the review, we briefly cover evidence for punishment in animals and argue that our focus on punishment in animal societies has eclipsed our appreciation for the important roles that reconciliation and third-party intervention play in reducing and resolving conflict across taxa.

Does punishment support social relationships in animals?

We do not wish to give the false impression that punishment is absent in animals. Indeed, punishment has been documented in the wild and under experimental contexts both within and between species (Box 4). For instance, within male–female pairs of cleaner fish (*Labroides dimidiatus*) who work together to clean client fish, males direct punishment (e.g., aggressive chases) toward cheating females who eat the more desirable client mucus as opposed to ‘cleaning’ the less desirable ectoparasites. In turn, females who have been punished are more cooperative – they clean more – in future interactions [133]. In laboratory experiments with chimpanzees, there is clear evidence for **second-party punishment** [134,135]: chimpanzees punish

those who have stolen from them and selectively engage in punishment in response to theft rather than loss of resources more generally. Moreover, recent work shows that chimpanzees who have been victims of a transgression (offered and then denied food) pay to watch the perpetrator receive punishment [136]. Notably, chimpanzees punish and pay to watch punishment only in second-party situations but not when they are third-party observers of transgressions [135,136].

Thus, punishment does happen in animals. However, its expression is rather limited both contextually and taxonomically. The examples outlined previously are some of the relatively few cases in which punishment has been clearly demonstrated outside of humans. Yet, conflict is ubiquitous in animal societies [137]. Thus, the ‘problem’ of animal conflict far outweighs the ‘solution’ of punishment in animals, raising the question: what other mechanisms are in place in animal societies to manage conflict?

Reconciliation

Following conflict, many animals engage in affiliative behaviors which appear to aid in reducing post-conflict stress and repairing relationships. These behaviors were first studied in primates [138], but have since been examined across a broad range of taxa [139–141]. Although several specific hypotheses have been put forth to explain reconciliation (see [140]), there is general consensus that reconciliation appears to play a role in sustaining valuable relationships [140,142].

Reconciliation in the wild is typically observed by examining affiliative behaviors after conflict and comparing them with rates of affiliative behaviors (within the same dyads) during control times (i.e., not after conflict). One aspect of the literature that is particularly striking is just how widespread reconciliation appears to be. To give a sense of the taxonomic breadth, reconciliation has been observed in goats (*Capra hircus* [143]), bottlenose dolphins (*Tursiops truncatus* [144]), and spotted hyenas (*Crocuta crocuta* [145]).

The kinds of affiliative behaviors involved in reconciliation show inter-species differences (Figure 1). Horses (*Equus caballus*) approach each other and spend time in close proximity to one

Box 4. Conflict and reconciliation between species

So far, the research covered in this review has centered on conflicts that occur within species. However, there are striking examples of conflict and, by extension, conflict management responses in between-species interactions. One example of between-species conflict that has emerged in recent years is observations of punishment within the context of octopus-fish hunting dyads. Octopus (*Octopus cyanea*) and fish of different species work together to flush out prey on reef systems and they mutually benefit from this arrangement because it increases their collective hunting success. When conflicts arise (e.g., the fish eats the prey), octopus have been known to deliver a swift punch to the fish, which is thought to serve as a partner control mechanism [193,194] (see Figure I).

Another example of between-species conflict can be found in the cleaner–client mutualism of *L. dimidiatus*. Here, cleaners cheat by eating the protective mucus covering client fish rather than the small parasites that they are ‘supposed’ to eat. Clients punish cleaners through aggressive chasing [195]. Alternatively, clients can terminate the relationship, a form of partner rejection. Within this context emerges a clear example of between-species reconciliation. Following cheating events, cleaner fish provide clients with ‘tactile massage’ using a pair of pectoral fins to massage the client’s body [196]. Tactile massage reduces stress in clients [197]. In addition to using tactile massage as a reconciliatory behavior, cleaner fish also seem to use it strategically: they are more likely to provide massage to predatory clients than to non-predatory clients [196].

Most recently, a study found that rats (*Rattus norvegicus*) reared with extensive human handling showed third-party intervention in conflicts involving unfamiliar mice (*Mus musculus*) [198]. Third-party intervention in these cases was preferentially directed toward aggressors and served to interrupt conflicts and reduce aggression.

Examples like this illustrate that conflict is pervasive in animal systems, occurring both within and between species. Reconciliatory behaviors may play a key role in managing conflict and stabilizing social relationships in these contexts.

**Trends in Cognitive Sciences**

Figure 1. Examples of between-species conflict and reconciliation. Top panel: octopus have been observed punching fish in response to conflicts in their collaborative feeding relationship (photo credit to Sandrine Rongère). Bottom panel: cleaner fish provide tactile massage to clients after cheating events (photo credit to Richard Ling).

another following conflict [146]. Red-necked wallabies (*Macropus rufogriseus*) affiliate after conflicts in different ways, including grooming and co-feeding [147]. Interestingly, they are more likely to reconcile after minor rather than severe conflict and, in this species, reconciliation leads to a reduction in both stress-related behavior and future conflict between dyads [147]. Wolves (*Canis*



Trends in Cognitive Sciences

Figure 1. Reconciliation post-conflict appears across a range of taxa. Reconciliation is measured in terms of the affiliative behaviors revealed following a conflict. Reconciliatory behaviors show species-specific patterns. Some illustrative examples include: red-necked wallabies engage in co-feeding (top left; Creative Commons license), chacma baboons produce acoustic calls called grunts (top right; photo credit to Pavel Špindler), horses approach each other and remain in close proximity (bottom left; photo credit to Tara Thissell), and wolves engage in social licking (bottom right; photo credit to Tambako The Jaguar).

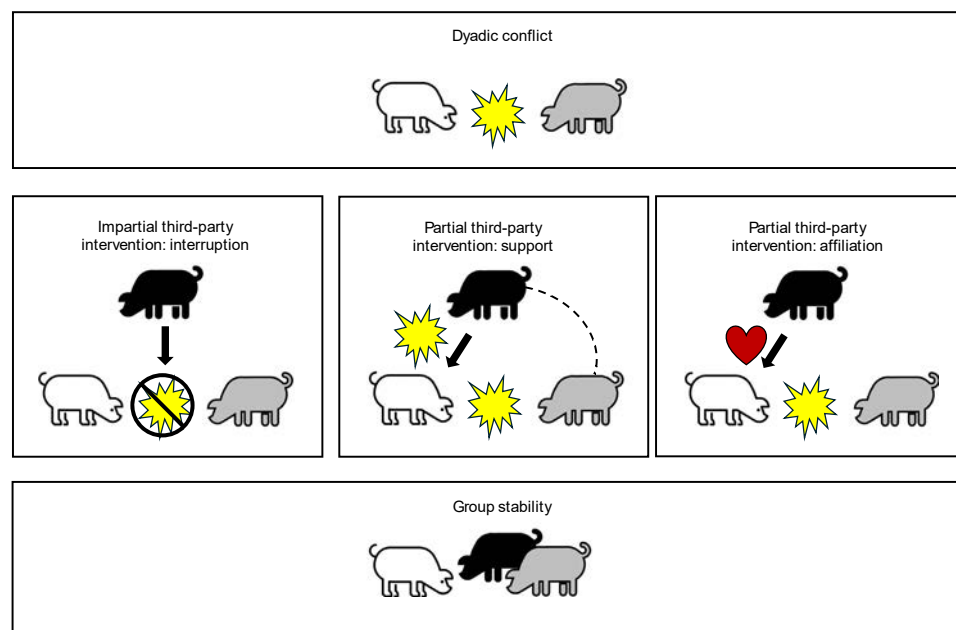
lupus) also reconcile after conflict by, for example, playing, sniffing, and licking each other, and these behaviors are initiated by both perpetrators and victims of aggression [148]. Similar findings have been reported in domestic dogs (*Canis familiaris* [149,150]), although dogs may engage in reconciliation less, or at least, do so in different ways, than their non-domesticated counterparts [151]. Finally, in chacma baboons (*Papio cynocephalus ursinus*), grunts are sufficient to reconcile conflict. In a playback experiment conducted in the wild, subordinates who had previously been attacked by a dominant individual who then grunted were less reactive to recorded screams from that dominant (potentially signaling a new attack) than those who had not heard a grunt from the previous attacker [152].

Third-party intervention

While clear examples of direct punishment from third parties have yet to be documented outside of humans [135,136], non-punitive intervention by third parties in the face of others' conflicts appears relatively common across taxa and appears to be directed toward reducing conflict (Figure 2).

One example of third-party intervention in animals is policing in social insects [153,154]. For instance, honeybee workers destroy eggs laid by other workers, which prevents the worker-laid eggs from developing into queens [155]. In the ant *Dinoponera quadricaps*, queens chemically mark subordinates who attempt to challenge her and subordinates are then attacked (and sometimes immobilized!) by other workers [156]. In these cases, policing confers inclusive fitness benefits to those who police by reducing reproductive conflict (allowing the queen to monopolize breeding opportunities) and so, indirectly, serves to promote stability of the colony.

Policing in primates appears more aligned – at the proximate level – with the goal of reducing conflict and restoring relationships. For instance, in pigtailed macaques (*Macaca nemestrina*), where policing is largely done by powerful individuals, third parties intervene in response to others' conflicts [157] and the experimental removal of policing individuals disrupts group stability [158]. In



Trends In Cognitive Sciences

Figure 2. Third parties intervene in dyadic conflict across a wide range of species. Intervention can be impartial, in which case neither the perpetrator nor the target of aggression is supported. Conversely, intervention can involve partial support, in which case the third party supports one of the two contestants (e.g., the more dominant of the two). Finally, intervention can be partial and affiliative, often directed toward the losing party in conflicts.

rhesus macaques (*Macaca mulatta*) [159], intervention by third parties has been shown to be prosocial in that it does not appear to confer any direct dominance benefits to the intervener. Work with captive chimpanzees has shown that policing is particularly common when groups are unstable [160], suggesting that this intervention strategy is particularly valuable in sustaining, or re-establishing, stable group dynamics.

Outside of behaviors that have specifically been labeled ‘policing’, a number of other forms of intervention by third parties post-conflict have been observed in animals. In all cases, third-party intervention appears to help stabilize groups. Savannah baboon (*Papio cynocephalus*) females – especially dominant females – regularly intervene in dyadic conflicts and typically support the more dominant of the two adversaries [161]. Intervention by third parties is also seen in sheep (*Ovis aries* [162]), where intervention frequency is positively related to rates of within-group aggression (i.e., third parties intervene more when conflict is common). In domestic dogs, third parties target the losing individual during littermate play, which has been argued to serve as practice for future agonistic interactions [163]. Playback experiments conducted with wild chimpanzees suggest that individuals expect intervention from the bond partners of their former opponents 2 h after fighting, underscoring the role that intervention by bystanders plays in regulating conflict over longer time scales [164]. Finally, recent work continues to highlight the taxonomic breadth of third-party intervention. In proboscis monkeys (*Nasalis larvatus*), vocal interventions by third-party males helps reduce conflict [165]. In pigs (*Sus scrofa*), third-party interveners participate in conflict and are more likely to win, which may eventually confer dominance benefits [166]. Third-party intervention can take the form of third-party affiliation [141]. For instance, high-ranking geladas (*Theropithecus gelada*) show third-party affiliation with low-ranking victims of aggression [167]. Similarly, in domestic dogs, third-party affiliation is directed toward victims of aggression [149]. Third-party affiliation has also been observed

in rooks (*Corvus frugilegus*) where, interestingly, reconciliation between opponents has not been observed [168].

A key take-home from the comparative work is that behaviors directed toward relationship restoration – both from those directly involved in conflict as well as from third parties – are widespread across animal societies. Taken together with work on humans, comparative data provide a richer context for the breadth and importance of response strategies that help mend fractured social relationships.

Concluding remarks

A full understanding of the psychology of justice in humans requires an appreciation of the myriad ways in which we respond to transgressions. Punishment has been in the spotlight for a long time and this has been for good reasons. Punishment is one mechanism that can help explain cooperation in large groups of non-relatives and it also forms the foundation of Western systems of justice. However, recent work has begun to expose cracks in the idea that punishment is the primary method by which humans seek justice. Instead, we often compensate victims, involve third-party mediators to help resolve conflict, and engage in the internal process of forgiving transgressors. These non-punitive responses, unlike many forms of punishment, center victims in the justice process (Boxes 5 and 6). Additionally, victim compensation, third-party mediation, and forgiveness can help restore justice following transgressions and help repair fractured relationships. Evidence from animal societies provides a broader context in which to situate these non-punitive responses. Whereas punishment in animal societies is relatively constrained in its expression and taxonomic distribution, response strategies that aid in repairing relationships – reconciliation and intervention by third parties – are widespread.

Despite these recent advances in our understanding of justice, many open questions remain (see Outstanding questions). For instance, it is not yet clear how different response strategies relate to each other, although there is some work in this area (e.g., [169]). Does victim compensation help foster forgiveness? Another open area is understanding whether non-punitive response strategies

Box 5. Bullying and justice

Bullying serves as a case study of where these justice-related responses carry important consequences in the real world. Existing research has explored both the interventions that effectively deter bullying in educational contexts as well as the ways in which children cope with peer victimization. Although much research has focused on policies and educational interventions that have the potential to reduce social exclusion and bullying (e.g., [199,200]), scholars agree that existing interventions have been only minimally effective in preventing bullying among children and adolescents [200,201]. As a result, a growing body of research has explored how teachers and school administrators can respond to bullying when it occurs.

Restorative justice approaches (which include peer and authority-led mediation, restorative conferences, community-building circles, and restorative conversations), when applied to incidents of bullying in schools, have been shown to be more effective than traditional punitive approaches in reducing rates of bullying, increasing rates of reports, and preventing acts of violence (for a review, see [202]). Restorative approaches have also been found to increase students' empathy and self-esteem relative to control conditions in experimental contexts [203]. Beyond these clear benefits for incidents of bullying, restorative justice approaches can also circumvent racial and gender biases that show up in punishment and zero-tolerance policies [202], further promoting justice in school environments.

Although restorative justice approaches can be effective in reducing bullying, it is still important to explore how bullied children can cope with their experiences in order to avoid negative psychological consequences and to disrupt cycles of peer victimization. Adolescents higher in forgiveness are less likely to act as bullies themselves and are also less likely to be victimized [204]. Forgiveness in response to bullying is correlated with lower rates of psychological symptoms, improved mental health and self-esteem, and lower rates of revenge-seeking [205]. Moreover, interventions to promote forgiveness among bullied children and adolescents have been found to effectively increase forgiveness, reduce anger, and increase empathy toward the transgressor [206].

Outstanding questions

What role does forgiveness play in small-scale societies? Work in anthropology has profoundly shaped our understanding of the role of restorative responses to transgressions across human societies. Yet, there has been relatively less work on forgiveness from an anthropological perspective. To what extent does forgiveness play a role in justice proceedings and relationship restoration across societies?

Do non-punitive responses promote behavioral reform? While punishment is known to lead to behavioral change – for instance, by promoting cooperative behaviors – it is less clear whether non-punitive, restorative responses have a similar effect on future behavior. Are victim compensation, mediation, and forgiveness as effective as punishment in deterring future transgressions?

Who should be the one to intervene? Social roles play an important role in guiding who responds to transgressions. Police officers in many Western societies have authority to intervene in conflict. In every-day justice contexts, however, these roles are less clear. Whose job is it to compensate victims? To help foster forgiveness? How do children acquire an understanding of these roles and integrate it into their emerging concepts of justice?

Do animals respond to norm violations? Many of the transgressions that crop up in human social interactions involve norms. For instance, one agent refuses to share with another, thereby violating a fairness norm. It is currently unclear whether animals have an understanding of norms. If they do, what are their responses to norm violations? If they do not, could this help explain why we see no evidence for third-party punishment which, in humans, is a signature of norm enforcement?

How are justice responses related? Victims of transgressions do not have to pursue just one response. For instance, a victim may seek compensation and punishment and, later, forgive the perpetrator. How are these responses related to one another and how do people decide when and how much to invest in the pursuit of each?

Box 6. Alternatives to punishment in criminal justice

The USA incarcerates more than 2 million of its citizens, more than any other country [207]. Yet a robust body of research suggests that incarceration is not effective in preventing or deterring crime and also has severe consequences for incarcerated individuals. Rates of recidivism for previously incarcerated individuals can be as high as 50% in the 3 years after release [208]. Several studies have found that early release from prison has no impact on rates of recidivism [209], suggesting that less time in prison does not lead to higher rates of recidivism. In addition, some experimental studies provide evidence that harsher punishments (i.e., longer sentences) may in fact increase recidivism [210], while at the same time having no significant deterrent effects [211]. Critics argue that incarceration is more expensive than other forms of punishment, fails to make communities safer, perpetuates racial and ethnic inequality, and generates harmful and severe consequences for offenders [207,212]. Moreover, research has documented that previously incarcerated individuals are less likely to hold jobs [213], have significantly higher suicide rates [214], and face social exclusion [215].

Although punishment decisions are often purportedly made with the goal of restoring justice for victims, victims themselves play relatively minor roles in criminal justice proceedings (with the exception of victim impact statements, which have been shown to influence sentencing decisions (e.g., [216])). Studies suggest that the majority of victims would prefer a focus on prevention and treatment over incarceration [212] and that engaging in restorative and forgiving interventions generate more positive outcomes for victims [217,218]. Moreover, the focus on punishing offenders still denies victims the opportunity to communicate with offenders, seek compensation, or advocate for structural changes. Other countries, including Finland, Denmark, and the UK, have incorporated more victim-oriented approaches to criminal justice proceedings, relying more heavily on community service programs, mediated justice procedures, and less severe punishment such as fines or constructive punishment [219,220]. These alternatives have been shown, in some research, to reduce reoffending [219] and to lead to more positive outcomes for offenders [212] and victims [217,218]. Together, this work points to the promise of restorative justice processes in helping reshape the US criminal justice system.

are as successful as punishment at reforming behavior and, again, some work has begun to explore this question (e.g., [114,170]). Additionally, it is crucial to expand the scope of empirical work on both non-punitive responses as well as on punishment beyond the current focus on WEIRD populations. Finally, given the many ways in which forgiveness is operationalized and measured (Box 3), it would be valuable for future work in this realm to start bridging methodological and conceptual gaps. Future work should address these, and other open questions, in an effort to round out our understanding of the different strategies that are used to manage conflict.

Our argument here is not that work on punishment should slow down. More work is undoubtedly needed to understand where and when it is pursued. Rather, we argue that recent developments across a range of disciplines have begun to paint a more complex picture of the psychology of justice in humans. Work with adults, children, different societies and different species suggests that non-punitive strategies are privileged in human and non-human animal societies. It is time for punishment to share the spotlight with these more restorative interventions.

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References

1. Clutton-Brock, T.H. and Parker, G.A. (1995) Punishment in animal societies. *Nature* 373, 209–216
2. Singh, M. and Boomsma, J.J. (2015) Policing and punishment across the domains of social evolution. *Oikos* 124, 971–982
3. Crockett, M.J. et al. (2014) The value of vengeance and the demand for deterrence. *J. Exp. Psychol. Gen.* 143, 2279–2286
4. Sarin, A. et al. (2021) Punishment is organized around principles of communicative inference. *Cognition* 208, 104544
5. Raihani, N.J. and Bshary, R. (2019) Punishment: one tool, many uses. *Evolut. Hum. Sci.* 1, e12
6. Tonry, M. (2009) Explanations of American punishment policies: a national history. *Punishment Soc.* 11, 377–394
7. Durkheim, É. et al. (1893) *The Division of Labor in Society*, New Free Press
8. Boyd, R. and Richerson, P.J. (1992) Punishment allows the evolution of cooperation (or anything else) in sizable groups. *Ethol. Sociobiol.* 13, 171–195

9. Bshary, R. and Bergmüller, R. (2008) Distinguishing four fundamental approaches to the evolution of helping. *J. Evol. Biol.* 21, 405–420
10. West, S.A. *et al.* (2007) Evolutionary explanations for cooperation. *Curr. Biol.* 17, R661–R672
11. Gülerk, Ö. *et al.* (2006) The competitive advantage of sanctioning institutions. *Science* 312, 108–111
12. Fehr, E. and Rockenbach, B. (2003) Detrimental effects of sanctions on human altruism. *Nature* 422, 137–140
13. Güth, W. *et al.* (1982) An experimental analysis of ultimatum bargaining. *J. Econ. Behav. Organ.* 3, 367–388
14. Fehr, E. and Fischbacher, U. (2004) Third-party punishment and social norms. *Evol. Hum. Behav.* 25, 63–87
15. Fehr, E. and Gächter, S. (2002) Altruistic punishment in humans. *Nature* 415, 137–140
16. Lohse, J. and Waichman, I. (2020) The effects of contemporaneous peer punishment on cooperation with the future. *Nat. Commun.* 11, 1815
17. Henrich, J. *et al.* (2006) Costly punishment across human societies. *Science* 312, 1767–1770
18. Henrich, J. *et al.* (2010) Markets, religion, community size, and the evolution of fairness and punishment. *Science* 327, 1480–1484
19. Mathew, S. and Boyd, R. (2011) Punishment sustains large-scale cooperation in prestate warfare. *Proc. Natl. Acad. Sci. U. S. A.* 108, 11375–11380
20. Kosfeld, M. and Rustagi, D. (2015) Leader punishment and cooperation in groups: experimental field evidence from commons management in Ethiopia. *Am. Econ. Rev.* 105, 747–783
21. Eriksson, K. *et al.* (2021) Perceptions of the appropriate response to norm violation in 57 societies. *Nat. Commun.* 12, 1481
22. Garfield, Z.H. *et al.* (2023) Norm violations and punishments across human societies. *Evol. Hum. Sci.* 5, e11
23. Molho, C. *et al.* (2024) Cross-societal variation in norm enforcement systems. *Philos. Trans. R. Soc. B* 379, 20230034
24. McAuliffe, K. *et al.* (2017) The developmental foundations of human fairness. *Nat. Hum. Behav.* 1, 0042
25. Marshall, J. and McAuliffe, K. (2022) Children as assessors and agents of third-party punishment. *Nat. Rev. Psychol.* 1, 334–344
26. Raihani, N.J. *et al.* (2012) Punishment and cooperation in nature. *Trends Ecol. Evol.* 27, 288–295
27. Pedersen, E.J. *et al.* (2018) The unresponsive avenger: more evidence that disinterested third parties do not punish altruistically. *J. Exp. Psychol. Gen.* 147, 514–544
28. Balafoutas, L. and Nikiforakis, N. (2012) Norm enforcement in the city: a natural field experiment. *Eur. Econ. Rev.* 56, 1773–1785
29. Balafoutas, L. *et al.* (2014) Direct and indirect punishment among strangers in the field. *Proc. Natl. Acad. Sci. U. S. A.* 111, 15924–15927
30. Balafoutas, L. *et al.* (2016) Altruistic punishment does not increase with the severity of norm violations in the field. *Nat. Commun.* 7, 13327
31. Balliet, D. *et al.* (2022) Prosocial and punishment behaviors in everyday life. *Curr. Opin. Psychol.* 43, 278–283
32. Molho, C. *et al.* (2020) Direct and indirect punishment of norm violations in daily life. *Nat. Commun.* 11, 3432
33. Hofmann, W. *et al.* (2018) Moral punishment in everyday life. *Personal. Soc. Psychol. Bull.* 44, 1697–1711
34. Pedersen, E.J. *et al.* (2020) When and why do third parties punish outside of the lab? A cross-cultural recall study. *Soc. Psychol. Personal. Sci.* 11, 846–853
35. Henrich, J. *et al.* (2010) The weirdest people in the world? *Behav. Brain Sci.* 33, 61–83
36. Van Doorn, J. *et al.* (2018) An exploration of third parties' preference for compensation over punishment: six experimental demonstrations. *Theor. Decis.* 85, 333–351
37. Van Doorn, J. and Brouwers, L. (2017) Third-party responses to injustice: a review on the preference for compensation. *Crime Psychol. Rev.* 3, 59–77
38. Philpot, R. *et al.* (2020) Would I be helped? Cross-national CCTV footage shows that intervention is the norm in public conflicts. *Am. Psychol.* 75, 66–75
39. Reverby, S.M. (2020) Compensation and reparations for victims and bystanders of the U.S. Public Health Service research studies in Tuskegee and Guatemala: who do we owe what? *Bioethics* 34, 893–898
40. Chavez, A.K. and Bicchieri, C. (2013) Third-party sanctioning and compensation behavior: findings from the ultimatum game. *J. Econ. Psychol.* 39, 268–277
41. Lotz, S. *et al.* (2011) Punitive versus compensatory reactions to injustice: emotional antecedents to third-party interventions. *J. Exp. Soc. Psychol.* 47, 477–480
42. FeldmanHall, O. *et al.* (2014) Fairness violations elicit greater punishment on behalf of another than for oneself. *Nat. Commun.* 5, 5306
43. Stallen, M. *et al.* (2018) Neurobiological mechanisms of responding to injustice. *J. Neurosci.* 38, 2944–2954
44. Bicchieri, C. and Maras, M. (2022) Intentionality matters for third-party punishment but not compensation in trust games. *J. Econ. Behav. Organ.* 197, 205–220
45. Leliveld, M.C. *et al.* (2012) Punishing and compensating others at your own expense: the role of empathic concern on reactions to distributive injustice. *Euro J. Social Psych.* 42, 135–140
46. Urbanska, K. *et al.* (2019) From injustice to action: the role of empathy and perceived fairness to address inequality via victim compensation. *J. Exp. Soc. Psychol.* 82, 129–140
47. Thulin, E.W. and Bicchieri, C. (2016) I'm so angry I could help you: moral outrage as a driver of victim compensation. *Soc. Philos. Policy* 32, 146–160
48. Adams, G.S. and Mullen, E. (2015) Punishing the perpetrator decreases compensation for victims. *Soc. Psychol. Personal. Sci.* 6, 31–38
49. Gromet, D.M. *et al.* (2012) A victim-centered approach to justice? Victim satisfaction effects on third-party punishments. *Law Hum. Behav.* 36, 375–389
50. Schmit, J.T. *et al.* (1988) Punitive damages: punishment or further compensation? *J. Risk Insurance* 55, 453
51. McAuliffe, K. *et al.* (2015) Costly third-party punishment in young children. *Cognition* 134, 1–10
52. Marshall, J. *et al.* (2021) Children punish third parties to satisfy both consequentialist and retributive motives. *Nat. Hum. Behav.* 5, 361–368
53. Arini, R.L. *et al.* (2021) Moral duty and equalization concerns motivate children's third-party punishment. *Dev. Psychol.* 57, 1325–1341
54. Lee, Y. and Warneken, F. (2022) Does third-party punishment in children aim at equality? *Dev. Psychol.* 58, 866–873
55. Ma, R. *et al.* (2023) Win-win or lose-lose: children prefer the form of equality. *Acta Psychol.* 246, 104251
56. Li, Y. *et al.* (2022) The squeaky wheel gets the grease: recipients' responses influence children's costly third-party punishment of unfairness. *J. Exp. Child Psychol.* 220, 105426
57. McAuliffe, K. and Dunham, Y. (2021) Children favor punishment over restoration. *Dev. Sci.* 24, e13093
58. Arini, R.L. *et al.* (2023) Children endorse deterrence motivations for third-party punishment but derive higher enjoyment from compensating victims. *J. Exp. Child Psychol.* 230, 105630
59. Lee, Y. and Warneken, F. (2020) Children's evaluations of third-party responses to unfairness: Children prefer helping over punishment. *Cognition* 205, 104374
60. Riedl, K. *et al.* (2015) Restorative justice in children. *Curr. Biol.* 25, 1731–1735
61. Zhou, Z. and Wong, W. (2022) Three- and five-year-olds' restorative intervention in moral transgressions. *Cogn. Dev.* 62, 101171
62. Yang, X. *et al.* (2021) Children's restorative justice in an intergroup context. *Soc. Dev.* 30, 663–683
63. Liu, X. *et al.* (2021) To punish or to restore: how children evaluate victims' responses to immorality. *Front. Psychol.* 12, 696160
64. Zhou, Z. and Wong, W. (2021) Young children's understanding of restorative justice. *Front. Psychol.* 12, 715279
65. Miller, D.T. and McCann, C.D. (1979) Children's reactions to the perpetrators and victims of injustices. *Child Dev.* 50, 861
66. Schwartz, F. and Chernyak, N. (2024) The good, the rich, and the powerful: how young children compensate victims of moral transgressions depending on moral character, wealth, and social dominance. *J. Exp. Child Psychol.* 247, 106045
67. Ejbye-Ernst, P. (2023) Does third-party intervention matter? A video-based analysis of the effect of third-party intervention on

- the continuation of interpersonal conflict behaviour. *Brit. J. Crim.* 63, 78–96
68. Halevy, N. and Halali, E. (2015) Selfish third parties act as peacemakers by transforming conflicts and promoting cooperation. *Proc. Natl. Acad. Sci. U. S. A.* 112, 6937–6942
 69. Wall, J.A. *et al.* (2001) Mediation: a current review and theory development. *J. Confl. Resolut.* 45, 370–391
 70. Hiltrop, J.M. (1988) Third-party mediation: some functional and strategic considerations. *Group Anal.* 21, 9–19
 71. Boyd, R. and Mathew, S. (2021) Arbitration supports reciprocity when there are frequent perception errors. *Nat. Hum. Behav.* 5, 596–603
 72. Burrell, N.A. *et al.* (2003) Evaluating peer mediation outcomes in educational settings: a meta-analytic review. *Confl. Resol. Q.* 21, 7–26
 73. Siddiqui, A. and Ross, H. (2004) Mediation as a method of parent intervention in children's disputes. *J. Fam. Psychol.* 18, 147–159
 74. Ross, H.S. and Lazinski, M.J. (2014) Parent mediation empowers sibling conflict resolution. *Early Educ. Dev.* 25, 259–275
 75. Wang, M. *et al.* (2020) The role of mediation in solving medical disputes in China. *BMC Health Serv. Res.* 20, 225
 76. He, T. *et al.* (2023) The role and development trend of third-party mediation in environmental disputes. *Sustainability* 15, 10197
 77. Clayton, G. and Dorussen, H. (2022) The effectiveness of mediation and peacekeeping for ending conflict. *J. Peace Res.* 59, 150–165
 78. Gartner, S.S. (2014) Third-party mediation of interstate conflicts: actors, strategies, selection, and bias. *Arbitr. Law Rev.* 6, 269–294
 79. Wiessner, P. (2020) The role of third parties in norm enforcement in customary courts among the Enga of Papua New Guinea. *Proc. Natl. Acad. Sci. U. S. A.* 117, 32320–32328
 80. Singh, M. and Garfield, Z.H. (2022) Evidence for third-party mediation but not punishment in Mentawai justice. *Nat. Hum. Behav.* 6, 930–940
 81. Fitouchi, L. and Singh, M. (2023) Punitive justice serves to restore reciprocal cooperation in three small-scale societies. *Evol. Hum. Behav.* 44, 502–514
 82. Glowacki, L. (2024) The evolution of peace. *Behav. Brain Sci.* 47, e1
 83. Garfield, Z.H. *et al.* (2020) Universal and variable leadership dimensions across human societies. *Evol. Hum. Behav.* 41, 397–414
 84. Garfield, Z.H. and Glowacki, L. (2023) Interpersonal conflicts and third-party mediation in a pastoralist society. *Evol. Hum. Behav.* 44, 613–623
 85. Rubin, J.Z. (1980) Experimental research on third-party intervention in conflict: toward some generalizations. *Psychol. Bull.* 87, 379–391
 86. Della Noce, D.J. (1999) Seeing theory in practice: an analysis of empathy in mediation. *Negot. J.* 15, 271–301
 87. Boca, S. *et al.* (2018) The effect of perspective taking on the mediation process. *Psychol. Res. Behav. Manag.* 411–416
 88. Gutenbrunner, L. and Wagner, U. (2016) Perspective-taking techniques in the mediation of intergroup conflict. *Peace Confl. J. Peace Psychol.* 22, 298–305
 89. Marshall, J. *et al.* (2024) The role of emotion in second- and third-party punishment. In *The Oxford Handbook of Evolution and the Emotions* (Al-Shawaf, L. and Shackelford, T.K., eds), pp. 997–1020, Oxford University Press
 90. Chaux, E. (2005) Role of third parties in conflicts among Colombian children and early adolescents. *Aggress. Behav.* 31, 40–55
 91. Burdelski, M. (2020) 'Say can I borrow it': teachers and children managing peer conflict in a Japanese preschool. *Linguist. Educ.* 59, 100728
 92. Johnson, D.W. and Johnson, R.T. (1996) Conflict resolution and peer mediation programs in elementary and secondary schools: a review of the research. *Rev. Educ. Res.* 66, 459–506
 93. Druckman, D. *et al.* (2021) Who is best at mediating a social conflict? Comparing robots, screens and humans. *Group Decis. Negot.* 30, 395–426
 94. Shen, S. *et al.* (2018) 'Stop. I see a conflict happening.': a robot mediator for young children's interpersonal conflict resolution. In *Proceedings of the 2018 ACM/IEEE International Conference on Human-Robot Interaction*, pp. 69–77, Association for Computing Machinery
 95. McCullough, M.E. (2001) Forgiveness: who does it and how do they do it? *Curr. Dir. Psychol. Sci.* 10, 194–197
 96. Fourie, M.M. *et al.* (2020) Parsing the components of forgiveness: psychological and neural mechanisms. *Neurosci. Biobehav. Rev.* 112, 437–451
 97. McCauley, T.G. *et al.* (2022) An evolutionary psychology view of forgiveness: individuals, groups, and culture. *Curr. Opin. Psychol.* 44, 275–280
 98. Brady, D.L. *et al.* (2023) Conceptualizing forgiveness: a review and path forward. *J. Organ. Behav.* 44, 261–296
 99. Yucel, M. and Vaish, A. (2021) Eliciting forgiveness. *WIREs. Cogn. Sci.* 12, e1572
 100. Van Der Wal, R.C. *et al.* (2017) Causes and consequences of children's forgiveness. *Child Dev. Perspect.* 11, 97–101
 101. Worthington, E.L. and Wade, N.G. (2019) A new perspective on forgiveness research. In *Handbook of Forgiveness* (2nd edn), Routledge
 102. Russell, L. (2020) The who, the what, and the how of forgiveness. *Philos Compass* 15, e12656
 103. Wamke, B. *et al.*, eds (2021) *Forgiveness and Its Moral Dimensions*, 1st edn Oxford University Press
 104. Potts, M.I. (2022) *Forgiveness an Alternative Account*, Yale University Press
 105. Worthington, E.L. *et al.* (2007) Forgiveness, health, and well-being: a review of evidence for emotional versus decisional forgiveness, dispositional forgiveness, and reduced unforgiveness. *J. Behav. Med.* 30, 291–302
 106. Fudenberg, D. *et al.* (2012) Slow to anger and fast to forgive: cooperation in an uncertain world. *Am. Econ. Rev.* 102, 720–749
 107. Van Houwelingen, G. *et al.* (2022) Two sides of the same coin: punishment and forgiveness in organizational contexts. *Front. Psychol.* 13, 908021
 108. Cornish, M.A. *et al.* (2020) Does empathy promotion necessarily lead to greater forgiveness? An experimental examination. *Curr. Psychol.* 39, 1001–1011
 109. Rizkalla, L. *et al.* (2008) The roles of emotion management and perspective taking in individuals' conflict management styles and disposition to forgive. *J. Res. Pers.* 42, 1594–1601
 110. Suzuki, M. and Jenkins, T. (2023) Apology–forgiveness cycle in restorative justice, but how? *Int. Rev. Victimol.* 29, 259–276
 111. McCullough, M.E. *et al.* (2014) Conciliatory gestures promote forgiveness and reduce anger in humans. *Proc. Natl. Acad. Sci. U. S. A.* 111, 11211–11216
 112. Ohtsubo, Y. *et al.* (2018) Costly apologies communicate conciliatory intention: an fMRI study on forgiveness in response to costly apologies. *Evol. Hum. Behav.* 39, 249–256
 113. Forster, D.E. *et al.* (2021) Experimental evidence that apologies promote forgiveness by communicating relationship value. *Sci. Rep.* 11, 13107
 114. Wallace, H.M. *et al.* (2008) Interpersonal consequences of forgiveness: does forgiveness deter or encourage repeat offenses? *J. Exp. Soc. Psychol.* 44, 453–460
 115. Gromet, D.M. and Okimoto, T.G. (2014) Back into the fold: the influence of offender amends and victim forgiveness on peer reintegration. *Bus. Ethics Q.* 24, 411–441
 116. Gollwitzer, M. and Okimoto, T.G. (2021) Downstream consequences of post-transgression responses: a motive-attribution framework. *Personal. Soc. Psychol. Rev.* 25, 275–294
 117. Hanke, K. and Vaclair, C.-M. (2016) Investigating the human value "forgiveness" across 30 countries: a cross-cultural meta-analytical approach. *Cross-Cult. Res.* 50, 215–230
 118. Joo, M. *et al.* (2019) How does culture shape conceptions of forgiveness? Evidence from Japan and the United States. *J. Cross-Cult. Psychol.* 50, 676–702
 119. Ho, M.Y. and Worthington, E.L. (2020) Is the concept of forgiveness universal? A cross-cultural perspective comparing western and eastern cultures. *Curr. Psychol.* 39, 1749–1756
 120. Enright, R.D. *et al.* (1989) The adolescent as forgiver. *J. Adolesc.* 12, 95–110
 121. Wainryb, C. *et al.* (2020) To err is human: forgiveness across childhood and adolescence. *Soc. Dev.* 29, 509–525

122. Amir, D. *et al.* (2021) Children are more forgiving of accidental harms across development. *J. Exp. Child Psychol.* 205, 105081
123. Amir, D. *et al.* (2024) The emergence of cooperative behaviors, norms, and strategies across five diverse societies. *PsyArXiv*, Published online April 18, 2024. <https://doi.org/10.31234/osf.io/7ud4>
124. Cushman, F. (2008) Crime and punishment: distinguishing the roles of causal and intentional analyses in moral judgment. *Cognition* 108, 353–380
125. McElroy, C.E. *et al.* (2023) Beyond accidents: young children's forgiveness of third-party intentional transgressors. *J. Exp. Child Psychol.* 228, 105607
126. Oostenbroek, J. and Vaish, A. (2019) The emergence of forgiveness in young children. *Child Dev.* 90, 1969–1986
127. Oostenbroek, J. and Vaish, A. (2019) The benefits of forgiving: young children respond positively to those who forgive. *J. Exp. Psychol. Gen.* 148, 1914–1924
128. Vaish, A. and Oostenbroek, J. (2022) Preferential forgiveness: the impact of group membership and remorse on preschoolers' forgiveness. *J. Exp. Psychol. Gen.* 151, 1132–1140
129. Mulvey, K.L. *et al.* (2022) The role of theory of mind, group membership, and apology in intergroup forgiveness among children and adolescents. *J. Exp. Psychol. Gen.* 151, 613–627
130. Karremans, J.C. *et al.* (2011) Maintaining harmony across the globe: the cross-cultural association between closeness and interpersonal forgiveness. *Soc. Psychol. Personal. Sci.* 2, 443–451
131. McLaughlin, A. *et al.* (2024) Developing conceptions of forgiveness across the lifespan. *Child Dev.* 95, 1915–1933
132. Smith, J.E. *et al.* (2016) Leadership in mammalian societies: emergence, distribution, power, and payoff. *Trends Ecol. Evol.* 31, 54–66
133. Raihani, N.J. *et al.* (2010) Punishers benefit from third-party punishment in fish. *Science* 327, 171
134. Jensen, K. *et al.* (2007) Chimpanzees are vengeful but not spiteful. *Proc. Natl. Acad. Sci. U. S. A.* 104, 13046–13050
135. Riedl, K. *et al.* (2012) No third-party punishment in chimpanzees. *Proc. Natl. Acad. Sci. U. S. A.* 109, 14824–14829
136. Mendes, N. *et al.* (2018) Preschool children and chimpanzees incur costs to watch punishment of antisocial others. *Nat. Hum. Behav.* 2, 45–51
137. Holekamp, K.E. and Strauss, E.D. (2016) Aggression and dominance: an interdisciplinary overview. *Curr. Opin. Behav. Sci.* 12, 44–51
138. De Waal, F.B.M. and Van Roosmalen, A. (1979) Reconciliation and consolation among chimpanzees. *Behav. Ecol. Sociobiol.* 5, 55–66
139. Silk, J.B. (2002) The form and function of reconciliation in primates. *Annu. Rev. Anthropol.* 31, 21–44
140. Arnold, K. and Aureli, F. (2007) Postconflict reconciliation. *Prim. Perspect.* 2, 608–625
141. Jensen, T.R. and Osvath, M. (2022) Post-conflict affiliation. In *Encyclopedia of Animal Cognition and Behavior* (Vonk, J. and Shackelford, T.K., eds), pp. 5463–5468. Springer International Publishing
142. Aureli, F. *et al.* (2002) Conflict resolution following aggression in gregarious animals: a predictive framework. *Anim. Behav.* 64, 325–343
143. Schino, G. (1998) Reconciliation in domestic goats. *Behaviour* 135, 343–356
144. Weaver, A. (2003) Conflict and reconciliation in captive bottlenose dolphins, *Tursiops truncatus*. *Mar. Mamm. Sci.* 19, 836–846
145. Wahaj, S.A. *et al.* (2001) Reconciliation in the spotted hyena (*Crocuta crocuta*). *Ethology* 107, 1057–1074
146. Cozzi, A. *et al.* (2010) Post-conflict friendly reunion in a permanent group of horses (*Equus caballus*). *Behav. Process.* 85, 185–190
147. Cordoni, G. and Norscia, I. (2014) Peace-making in marsupials: the first study in the red-necked wallaby (*Macropus rufogriseus*). *PLoS One* 9, e86859
148. Cordoni, G. and Palagi, E. (2008) Reconciliation in wolves (*Canis lupus*): new evidence for a comparative perspective. *Ethology* 114, 298–308
149. Cools, A.K.A. *et al.* (2008) Canine reconciliation and third-party-initiated postconflict affiliation: do peacemaking social mechanisms in dogs rival those of higher primates? *Ethology* 114, 53–63
150. Walters, K.A.F. *et al.* (2020) Reconciliation in domestic dogs (*Canis familiaris*): evidence for the uncertainty reduction hypothesis. *Appl. Anim. Behav. Sci.* 226, 104987
151. Cafazzo, S. *et al.* (2018) The effect of domestication on post-conflict management: wolves reconcile while dogs avoid each other. *R. Soc. Open Sci.* 5, 171553
152. Cheney, D.L. *et al.* (1995) The role of grunts in reconciling opponents and facilitating interactions among adult female baboons. *Anim. Behav.* 50, 249–257
153. Ratnieks, F.L.W. and Wenseleers, T. (2005) Policing insect societies. *Science* 307, 54–56
154. Ratnieks, F. and Wenseleers, T. (2008) Altruism in insect societies and beyond: voluntary or enforced? *Trends Ecol. Evol.* 23, 45–52
155. Ratnieks, F.L.W. and Visscher, P.K. (1989) Worker policing in the honeybee. *Nature* 342, 796–797
156. Monnin, T. *et al.* (2002) Pretender punishment induced by chemical signalling in a queenless ant. *Nature* 419, 61–65
157. Flack, J.C. *et al.* (2005) Social structure, robustness, and policing cost in a cognitively sophisticated species. *Am. Nat.* 165, E126–E139
158. Flack, J.C. *et al.* (2006) Policing stabilizes construction of social niches in primates. *Nature* 439, 426–429
159. Beisner, B.A. and McCowan, B. (2013) Policing in nonhuman primates: partial interventions serve a prosocial conflict management function in rhesus macaques. *PLoS One* 8, e77369
160. Von Rohr, C.R. *et al.* (2012) Impartial third-party interventions in captive chimpanzees: a reflection of community concern. *PLoS One* 7, e32494
161. Silk, J.B. *et al.* (2004) Patterns of coalition formation by adult female baboons in Amboseli, Kenya. *Anim. Behav.* 67, 573–582
162. Papadaki, K. *et al.* (2022) Do sheep (*Ovis aries*) perform third-party interventions? *Ethology* 128, 657–667
163. Ward, C. *et al.* (2009) Third-party interventions in dyadic play between littermates of domestic dogs, *Canis lupus familiaris*. *Anim. Behav.* 78, 1153–1160
164. Wittig, R.M. *et al.* (2014) Triadic social interactions operate across time: a field experiment with wild chimpanzees. *Proc. R. Soc. B* 281, 20133155
165. Scheumann, M. *et al.* (2022) Third-party vocal intervention in the proboscis monkey (*Nasalis larvatus*). *Int. J. Primatol.* 43, 698–711
166. Maffezzini, N. *et al.* (2023) Third-party intervention and post-conflict behaviour in agonistic encounters of pigs (*Sus scrofa*). *Front. Zool.* 20, 28
167. Palagi, E. *et al.* (2018) High-ranking geladas protect and comfort others after conflicts. *Sci. Rep.* 8, 15291
168. Seed, A.M. *et al.* (2007) Postconflict third-party affiliation in rooks, *Corvus frugilegus*. *Curr. Biol.* 17, 152–158
169. Komiya, A. *et al.* (2018) Providing compensation promotes forgiveness for replaceable, but not irreplaceable, losses. *J. Soc. Psychol.* 158, 680–693
170. McNulty, J.K. and Russell, V.M. (2016) Forgive and forget, or forgive and regret? Whether forgiveness leads to less or more offending depends on offender agreeableness. *Personal. Soc. Psychol. Bull.* 42, 616–631
171. Barclay, P. (2016) Biological markets and the effects of partner choice on cooperation and friendship. *Curr. Opin. Psychol.* 7, 33–38
172. Martin, J. *et al.* (2019) The psychology of partner choice. *PsyArXiv*, Published online September 18, 2019. <https://doi.org/10.31234/osf.io/ueqhz>
173. Martin, J.W. and Cushman, F. (2015) To punish or to leave: distinct cognitive processes underlie partner control and partner choice behaviors. *PLoS One* 10, e0125193
174. Martin, J.W. *et al.* (2022) An earlier role for intent in children's partner choice versus punishment. *J. Exp. Psychol. Gen.* 151, 597–612
175. Loustau, T. *et al.* (2024) The impact of group membership on punishment versus partner rejection. *Sci. Rep.* 14, 22238
176. Schino, G. and Aureli, F. (2017) Reciprocity in group-living animals: partner control versus partner choice. *Biol. Rev.* 92, 665–672

177. Raihani, N.J. and Bshary, R. (2015) Third-party punishers are rewarded, but third-party helpers even more so. *Evolution* 69, 993–1003
178. Jordan, J.J. *et al.* (2016) Third-party punishment as a costly signal of trustworthiness. *Nature* 530, 473–476
179. Dhaliwal, N.A. *et al.* (2021) Reputational and cooperative benefits of third-party compensation. *Organ. Behav. Hum. Decis. Process.* 164, 27–51
180. Adams, G.S. and Mullen, E. (2013) Increased voting for candidates who compensate victims rather than punish offenders. *Soc. Justice Res* 26, 168–192
181. Heffner, J. and FeldmanHall, O. (2019) Why we don't always punish: preferences for non-punitive responses to moral violations. *Sci. Rep.* 9, 13219
182. DiDonato, T.E. *et al.* (2015) The fallout of forgiveness: how forgiveness predicts third-party perceptions of the forgiver and the forgiver's relationships. *J. Soc. Pers. Relat.* 32, 417–440
183. Forster, D.E. *et al.* (2020) Forgiveness takes place on an attitudinal continuum from hostility to friendliness: toward a closer union of forgiveness theory and measurement. *J. Pers. Soc. Psychol.* 119, 861–880
184. Pearce, H. *et al.* (2018) The barriers to forgiveness scale: a measure of active and reactive reasons for withholding forgiveness. *Personal. Individ. Differ.* 134, 337–347
185. Wenzel, M. *et al.* (2021) Dynamics of moral repair: forgiveness, self-forgiveness, and the restoration of value consensus as interdependent processes. *Personal. Soc. Psychol. Bull.* 47, 607–626
186. Fehr, R. *et al.* (2010) The road to forgiveness: a meta-analytic synthesis of its situational and dispositional correlates. *Psychol. Bull.* 136, 894–914
187. Zechmeister, J.S. and Romero, C. (2002) Victim and offender accounts of interpersonal conflict: autobiographical narratives of forgiveness and unforgiveness. *J. Pers. Soc. Psychol.* 82, 675–686
188. Will, G.-J. *et al.* (2015) Acting on social exclusion: neural correlates of punishment and forgiveness of excluders. *Soc. Cogn. Affect. Neurosci.* 10, 209–218
189. Martin, J.W. and Cushman, F. (2016) Why we forgive what can't be controlled. *Cognition* 147, 133–143
190. McCullough, M.E. *et al.* (2013) Cognitive systems for revenge and forgiveness. *Behav. Brain Sci.* 36, 1–15
191. Strelnan, P. (2018) Justice and forgiveness in interpersonal relationships. *Curr. Dir. Psychol. Sci.* 27, 20–24
192. Wenzel, M. and Okimoto, T.G. (2014) On the relationship between justice and forgiveness: are all forms of justice made equal? *Br. J. Soc. Psychol.* 53, 463–483
193. Sampaio, E. *et al.* (2021) Octopuses punch fishes during collaborative interspecific hunting events. *Ecology* 102, 1–4
194. Sampaio, E. *et al.* (2024) Multidimensional social influence drives leadership and composition-dependent success in octopus–fish hunting groups. *Nat. Ecol. Evol.* 14, 22238
195. Bshary, R. and Grutter, A.S. (2005) Punishment and partner switching cause cooperative behaviour in a cleaning mutualism. *Biol. Lett.* 1, 396–399
196. Bshary, R. and Würth, M. (2001) Cleaner fish *Labroides dimidiatus* manipulate client reef fish by providing tactile stimulation. *Proc. R. Soc. Lond. B* 268, 1495–1501
197. Soares, M.C. *et al.* (2011) Tactile stimulation lowers stress in fish. *Nat. Commun.* 2, 534
198. Mikami, K. *et al.* (2024) Third-party punishment-like behavior in a rat model. *Sci. Rep.* 14, 22310
199. Gregus, S.J. *et al.* (2020) Toward evidence-based interventions for chronically bullied children: candidate mechanisms and potential strategies. *Evid.-Based Pract. Child Adolesc. Mental Health* 5, 83–101
200. Swearer, S.M. *et al.* (2017) Bullying intervention in adolescence: the intersection of legislation, policies, and behavioral change. *Adolesc. Res. Rev.* 2, 23–35
201. Merrell, K.W. *et al.* (2008) How effective are school bullying intervention programs? A meta-analysis of intervention research. *Sch. Psychol. Q.* 23, 26–42
202. Lodi, E. *et al.* (2021) Use of restorative justice and restorative practices at school: a systematic literature review. *Int. J. Environ. Res. Public Health* 19, 96
203. Wong, D.S. *et al.* (2010) Program effectiveness of whole-school approach for tackling bullying in Hong Kong primary schools. *Ed. Res. J.* 25, 103–124
204. Quintana-Orts, C. *et al.* (2021) The relationship between forgiveness, bullying, and cyberbullying in adolescence: a systematic review. *Trauma Violence Abuse* 22, 588–604
205. Flanagan, K.S. *et al.* (2012) The potential of forgiveness as a response for coping with negative peer experiences. *J. Adolesc.* 35, 1215–1223
206. Rapp, H. *et al.* (2022) A meta-analysis of forgiveness education interventions' effects on forgiveness and anger in children and adolescents. *Child Dev.* 93, 1249–1269
207. Corbett, H. (2023) Why alternatives to incarceration are good for communities, workplaces and the economy. *Forbes*, 24 February
208. Antenangeli, L. and Durose, M.R. (2021) Recidivism of prisoners released in 24 states in 2008: a 10-year follow-up period (2008–2018). In *US Department of Justice, Office of Justice Programs, Bureau of Justice Statistics*, US Department of Justice
209. Mauer, M. and Ghandnoosh, N. (2014) *Fewer Prisoners, Less Crime: A Tale of Three States*, Policy Brief, The Sentencing Project
210. Chen, M.K. and Shapiro, J.M. (2007) Do harsher prison conditions reduce recidivism? A discontinuity-based approach. *Am. Law Econ. Rev.* 9, 1–29
211. Bun, M.J.G. *et al.* (2020) Crime, deterrence and punishment revisited. *Empir. Econ.* 59, 2303–2333
212. Washington, M. (2021) *Beyond jails: community-based strategies for public safety*, Vera Institute of Justice
213. Carson, A. *et al.* (2021) Employment of persons released from federal prison in 2010. In *US Department of Justice, Office of Justice Programs, Bureau of Justice Statistics*, US Department of Justice
214. Morgan, E.R. *et al.* (2022) Incarceration and subsequent risk of suicide: a statewide cohort study. *Suicide Life Threat. Behav.* 52, 467–477
215. Brew, B. *et al.* (2022) Sticky stigma: the impact of incarceration on perceptions of personality traits and deservingness. *Soc. Forces* 100, 1910–1934
216. Dufour, G.K. *et al.* (2023) The relationship between victim impact statements and judicial decision making: an archival analysis of sentencing outcomes. *Law Hum. Behav.* 47, 484–498
217. Witvliet, C.V.O. *et al.* (2008) Retributive justice, restorative justice, and forgiveness: an experimental psychophysiology analysis. *J. Exp. Soc. Psychol.* 44, 10–25
218. Nascimento, A.M. *et al.* (2023) The psychological impact of restorative justice practices on victims of crimes—a systematic review. *Trauma Violence Abuse* 24, 1929–1947
219. Wenzel, M. *et al.* (2008) Retributive and restorative justice. *Law Hum. Behav.* 32, 375–389
220. Newton, A. (1981) Alternatives to imprisonment - an international perspective. *Crim. Just. Abstr.* 13, 134–148